

Exercise Sheet No. 5

Issued: 2025-11-17
Due: 2025-11-24
09:00 hrs

Strings and Typing

Note: Submitting your work will not earn you any bonus points. Your tutor will provide you with brief feedback on your submission. This exercise is therefore intended as practice and a way for you to check your learning progress!

You may use anything you like for this sheet, but simply copying and uploading automatically generated solutions (e.g. via an LLM) is not helpful. The sources must be cited!

At the beginning of all your source code files (.py), specify the variable `__author__` (at the beginning of the source code) – this is good practice, as it will be important for the code review tasks that are relevant to your coursework: `__author__ = "<Matr-Nr>, <Nachname>"`

Digital submissions that are not in .pdf or .txt format for texts or .py for code will not be reviewed. If submitting multiple files, they must be combined into a .zip file.

Whenever we refer to Python programmes, we mean Python code that is executable with at least Python version 3.10. Please work completely without imports at this time.

Please check the rules document in the Moodle course for further details on bonuspoints: <https://moodle.studiumdigitale.uni-frankfurt.de/moodle/mod/resource/view.php?id=323922>

(currently only in German, but we are sure you can translate it with online tools if needed).

Exercise 1: String Slicing

From the string `"I'm as shocked as you are"`, produce the following outputs, each with only a single slicing expression:

- | | |
|------------------------------|---|
| a) <code>"r_o_adkoss"</code> | d) <code>"mse_"</code> |
| b) <code>"shocked"</code> | e) The 2nd-, 5th-, 8th-, and 11th-to-last elements. |
| c) <code>"era_uoy"</code> | f) All elements starting with the 7th. |

Exercise 2: Casting and Coercion

In your own words, explain the difference between casting and coercion. Give an example of each and explain specifically where casting or coercion occurs.

Exercise 3: String Functions

Write a Python 3 program that receives a string from the console and sequentially prints the following info to the console:

- a) the string's length,
- b) whether the string contains only letters,
- c) whether the string is a valid Python identifier,
- d) whether the string ends in `ing`,
- e) the string in lowercase letters,
- f) the string with all occurrences of `a` replaced by `e`.

Additionally, specify three different test cases for your program. No additional documentation is necessary.

First, familiarize yourself with the string functions: <https://docs.python.org/3/library/stdtypes.html#text-sequence-type-str>. The use of loops is not permitted in this exercise.

Exercise 4: The Needle in the Haystack

Write a Python 3 program that replicates the functionality of `str.find()`.

Your program should first read the two strings *needle* and *haystack* from the console, and the output the lowest index at which the string *needle* occurs within the string *haystack*. If *needle* is not found in *haystack*, `-1` shall be output.

Additionally, specify three test cases for your program, and write a short `README` file explaining how to use your program.

Use only loops, slicing, and indexing for your implementation. The use of string functions like `str.find()`, `str.index()`, `str.split()`, etc. is not permitted in this exercise.